Comparing stock error with different stock pricing models

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1 Introduction

This paper aims to delve into the differing types of stock pricing strategies and models used in quantitative trading and compare them using an investigation in MS Excel. The three strategies covered in this paper are Geometric Brownian Motion (GBM), Ornstein–Uhlenbeck Mean Reversion (OUMR), and Cox–Ingersoll–Ross (CIR). The rest, namely Variance Gamma (VG), Heston Stochastic Volatility (HSV), and GARCH(1,1) and its variants, will be explored in a separate paper.

The structure of this paper will roughly follow that of the previous paper on testing shock functions, except there will not be as many derivations, as solving stochastic differential equations (SDEs) is currently outside scope. The paper uses 2 years of historical data to predict the stock price 11 days ahead. The predictions of the models are then compared against the actual prices, the percentage error is found, and each model is compared based on the statistical measures of error produced in their Monte Carlo simulations, to conclude on the accuracy and precision of each model. Penny stocks were initially considered as a third category for comparison, but were excluded from the investigation on the grounds that their historical data is too inconsistent and, in many cases, does not stretch back the full 2 years required by the model, making reliable comparison impossible.

For instructions on how to use the Excel model, please refer to Section 8 - Instructions for use.

2 Assumptions

The investigation assumes that returns are normally distributed — which, as described in a related paper published with UNSW's Parabola (linked at the end of this section, is not strictly correct. This assumption is made here for two reasons: first, deriving an analytical closed form of the inverse c.d.f. I derived is mathematically impossible, and

computer-aided numerical approximations in Excel are currently out of my scope; second, it keeps the investigation consistent, since `norm.s.inv()` is the most commonly used shock function in quantitative backtesting, allowing the effects of each prediction model to be assessed more clearly and reliably against a common baseline.

Link to paper (you need to download this report for it to work):Weighting for a better distribution - Gram-Charlier in the world of finance

3 Meet our models

3.1 Geometric Brownian Motion (GBM)

The simplest model used, Geometric Brownian Motion, consists of two terms: a drift term proportional to time, and a volatility term assumed to be normally distributed. It originally came from molecular physics, where it was used to model the diffusion of a chemical in a solvent at molecular scales. It satisfies the SDE:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t \,,$$

which upon discretisation turns into:

$$S_{t+1} = S_t \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z\right],$$

where:

- S_t = value of stock at time t,
- μ = drift of stock (the predictable part),
- σ = volatility of stock (the unpredictable part),
- dW_t = increment of a standard Wiener process ($dW_t \sim N(0, dt)$),
- Z = discretised version of dW_t ($Z \sim N(0, 1)$), evaluated using `norm.s.inv()`.

GBM is used primarily for Monte Carlo simulations, where the small errors caused by assuming normally distributed returns iron themselves out when simulated thousands of times. It is also computationally easy to simulate when compared to other models, and so it is well suited for options pricing using the Black–Scholes model. There are better and more sophisticated price models, so it is not used as much for short-term price predictions. One of the main reasons for this is that it assumes markets are inefficient and does not account for arbitrage.

3.2 Ornstein–Uhlenbeck Mean Reversion (OUMR)

OUMR is used on prices that are mean reverting, such as commodities (oil, gold, silver, platinum, and so on). It is not commonly used to model stock price changes, but it is included here as a method of comparison to assess its performance against better models. OUMR was one of the first pricing strategies used by Jim Simons and Renaissance Technologies to profit from commodity trading, and although the market has changed significantly since then, it still has its place in pair trading. As the name suggests, it assumes that prices will revert closer to their mean, with a strength proportional to the difference between the running average and the current price.

This behaviour vaguely resembles the dampened simple harmonic motion observed in spring oscillators, with the addition of a volatility shock function. It was initially developed as a way to describe the motion of a Brownian particle with friction.

The SDE of OUMR is:

$$dS_t = \theta(\bar{S} - S_t)\, dt + \sigma\, dW_t\,,$$

which, when discretised using Euler–Maruyama, turns into:

$$S_{t+1} = S_t + \theta(\bar{S} - S_t)\, \Delta t + \sigma\sqrt{\Delta t}\, Z\,,$$

where:

- S_t = value of stock at time t,
- θ = strength of mean reversion ($\theta > 0$),
- $\bar{S}$ = mean value of stock (running average),
- σ = volatility of stock,
- dW_t = standard Wiener increment ($dW_t \sim N(0, dt)$),
- Z = discretised version of dW_t ($Z \sim N(0, 1)$), evaluated using `norm.s.inv()`.

3.3 Cox–Ingersoll–Ross (CIR)

CIR is a close cousin of OUMR, with a couple of characteristic differences. In the other models, it is theoretically possible for prices to go negative, but in CIR the volatility term is proportional to the square root of the current stock price. This means that as the stock price approaches 0, the size of the random jumps decreases, becoming nearly undetectable. This feature ensures that the stock price never goes below 0, provided Feller's condition holds, and makes the size of the jumps proportional to the current price — an accurate reflection of how the sizes of random jumps change as the price of the stock varies. CIR is commonly used for interest rates and credit spreads.

The SDE of CIR is:

$$dS_t = \theta(\bar{S} - S_t)\, dt + \sigma\sqrt{S_t}\, dW_t\,,$$

where Feller's condition is:

$$2\theta\bar{S} > \sigma^2 \,,$$

which when discretised turns into:

$$S_{t+1} = S_t + \theta(\bar{S} - S_t)\,\Delta t + \sigma\sqrt{\Delta t}\,Z \,,$$

where:

- S_t = value of stock at time t,
- θ = strength of mean reversion ($\theta > 0$),
- $\bar{S}$ = mean value of stock (running average),
- σ = volatility of stock,
- dW_t = standard Wiener increment ($dW_t \sim N(0, dt)$),
- $Z = \max(S_t, 0)$ (preferred) or $|S_t|$, used to enforce non-negativity of the stock price in discretisation.

4 Method and calculations

The following is the method behind the process of using historical data to predict the price for the next 11 days, and comparing it to the actual price for those 11 days.

(a) Two years of historical data is pulled from MS Excel using the `stockhistory()` function for a specific company. Data is from 1/1/24 to 1/1/26 and is pulled in daily intervals (date and close price used).

(b) The standard return and log returns are generated in the nearby columns.

(c) The return metrics and necessary values for the models are calculated.

(d) The historical data for the next 11 trading days are pulled for comparison.

(e) The prices of each model are predicted, and the average historical delta (percentage error) is found using:

$$\text{Abs \% error} = \frac{|\,\text{Predicted price} - \text{Historical price}\,|}{\text{Historical price}} \times 100\% \,.$$

(f) Step (e) is repeated 500 times for each model, using a Monte Carlo simulation.

(g) The 500 values' means and standard deviations are found and graphed next to each other.

5 Results and graphs

The companies' results graphs are segregated by industry to assess each model's performance using their statistical metrics and relative to each other, in order to assess which model would be the most appropriate to use, and by how much it outperforms the others.

Blue-Chip Stocks

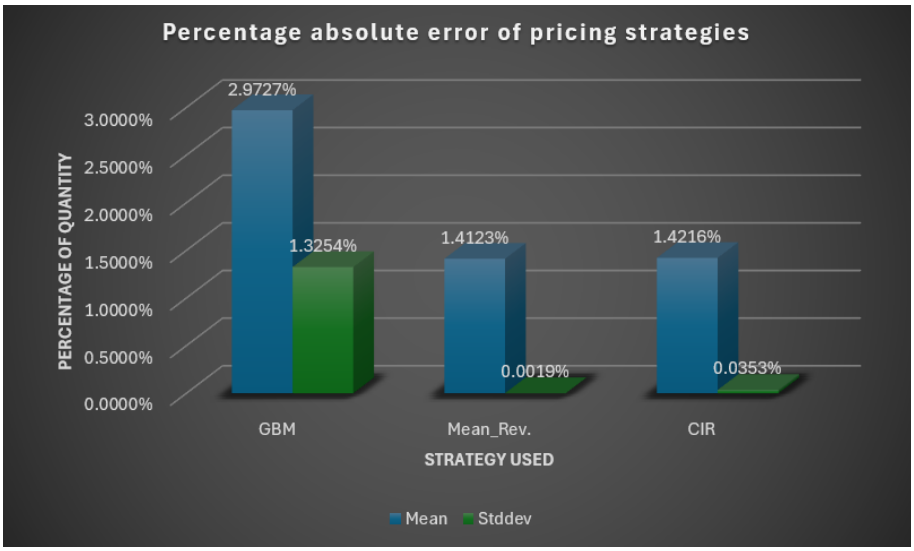

Figure 1: **MSFT** prediction error metrics

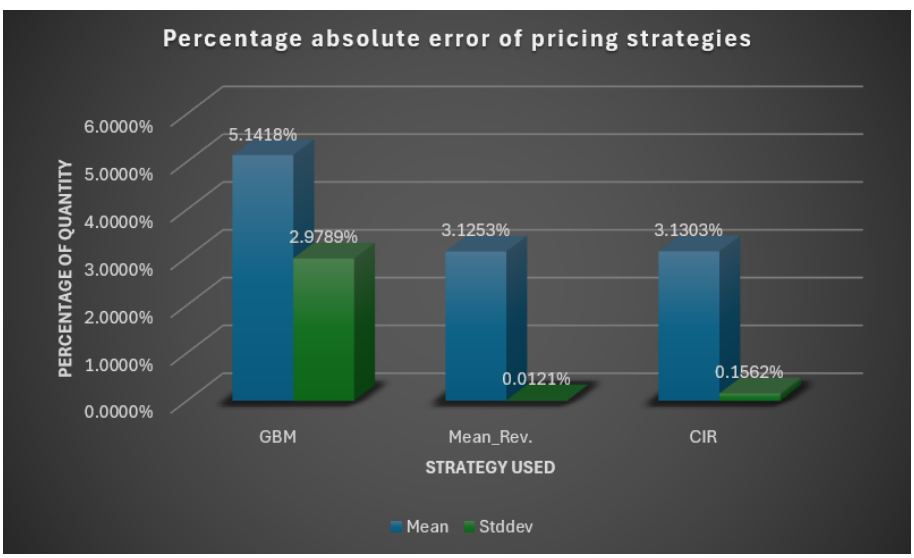

Figure 2: **AAPL** prediction error metrics

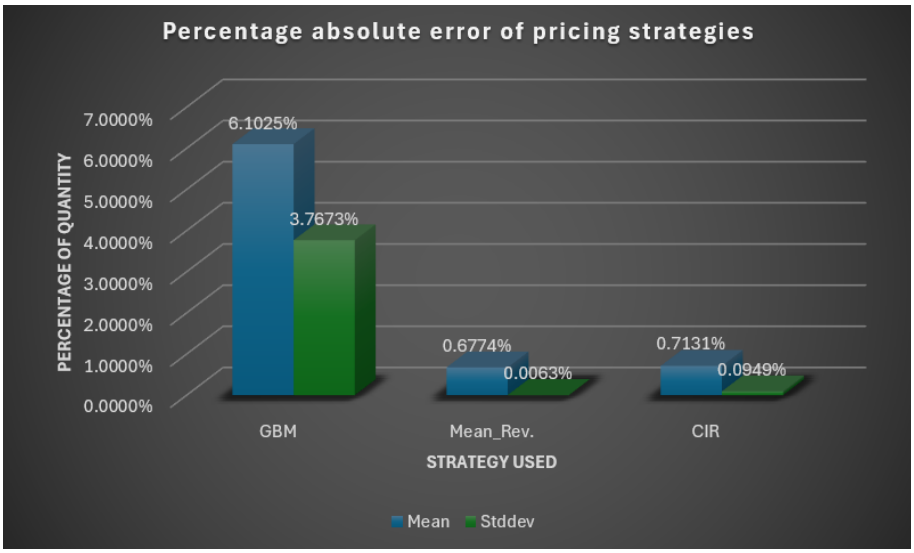

Figure 3: **NVDA** prediction error metrics

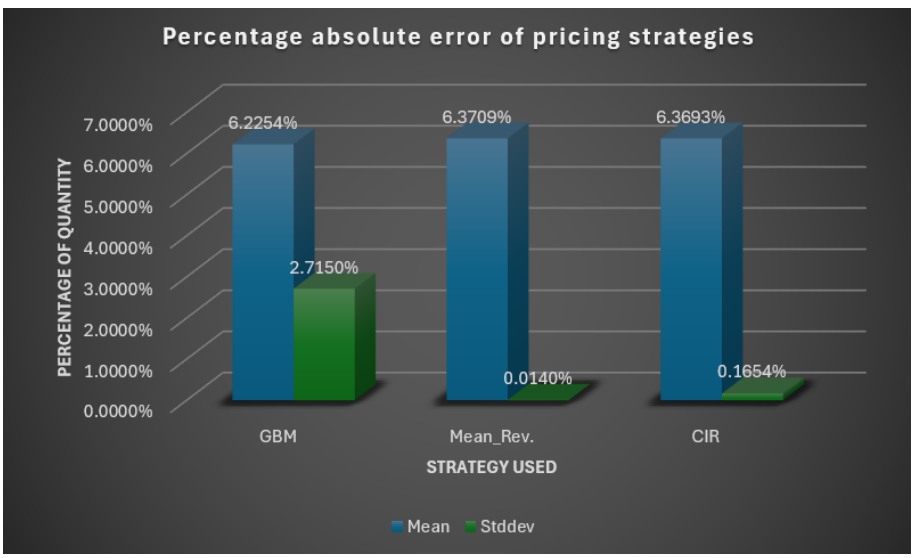

Figure 4: **AMZN** prediction error metrics

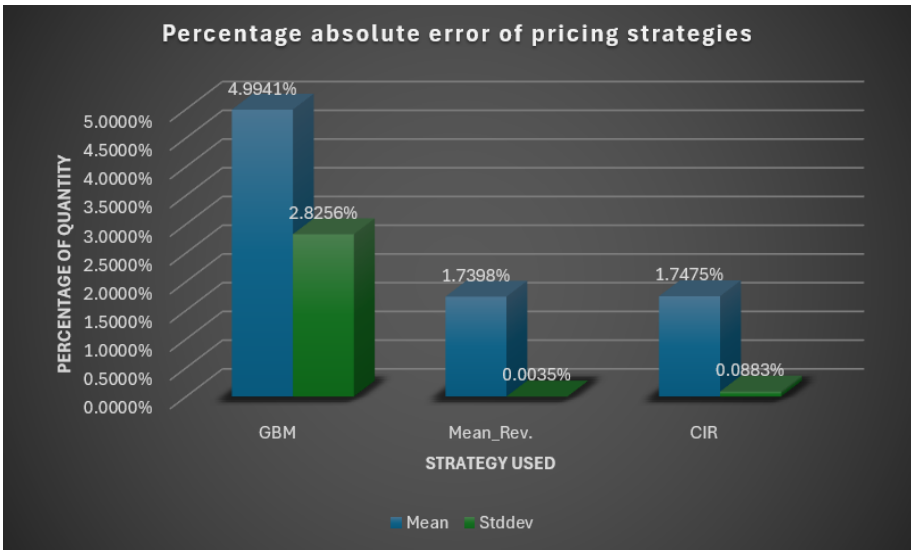

Figure 5: **META** prediction error metrics

Commodity-Based Stocks

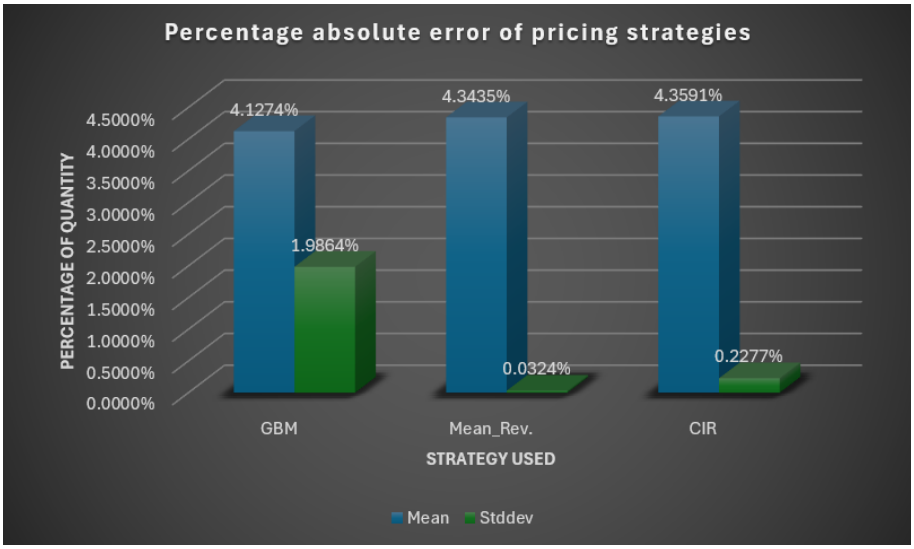

Figure 6: **RIO(NYSE)** prediction error metrics

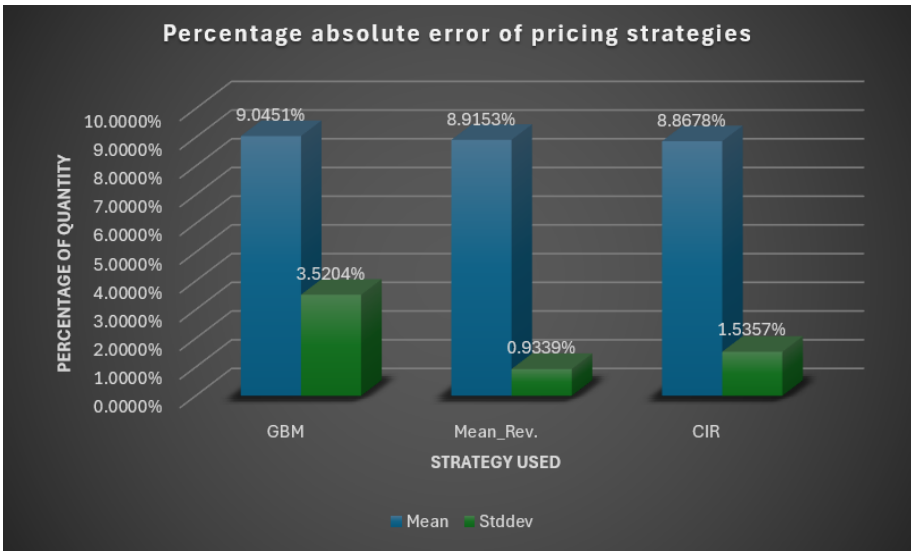

Figure 7: **S32** prediction error metrics

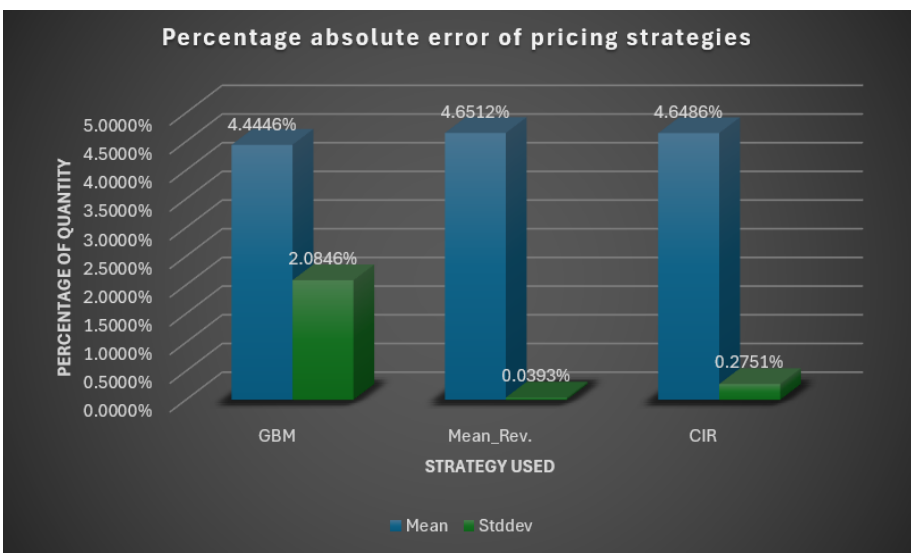

Figure 8: **BHP(NYSE)** prediction error metrics

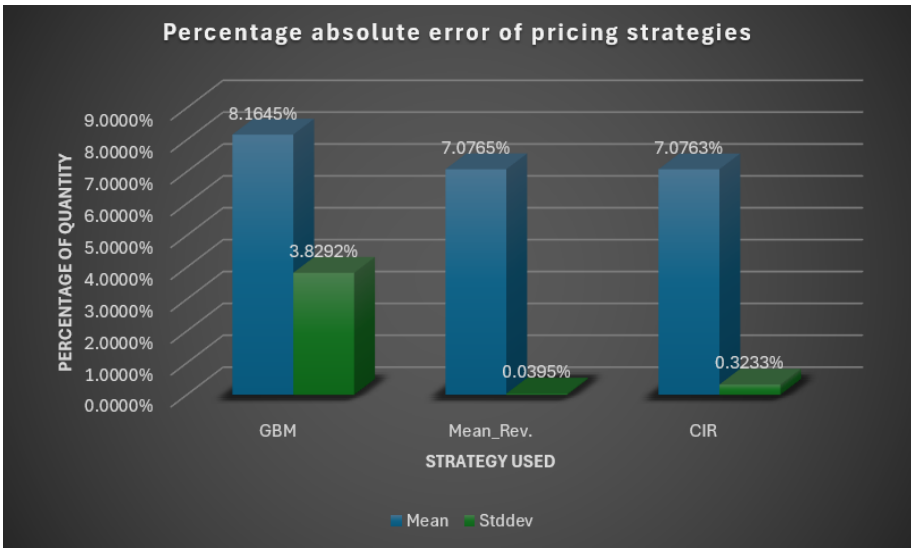

Figure 9: **NEM** prediction error metrics

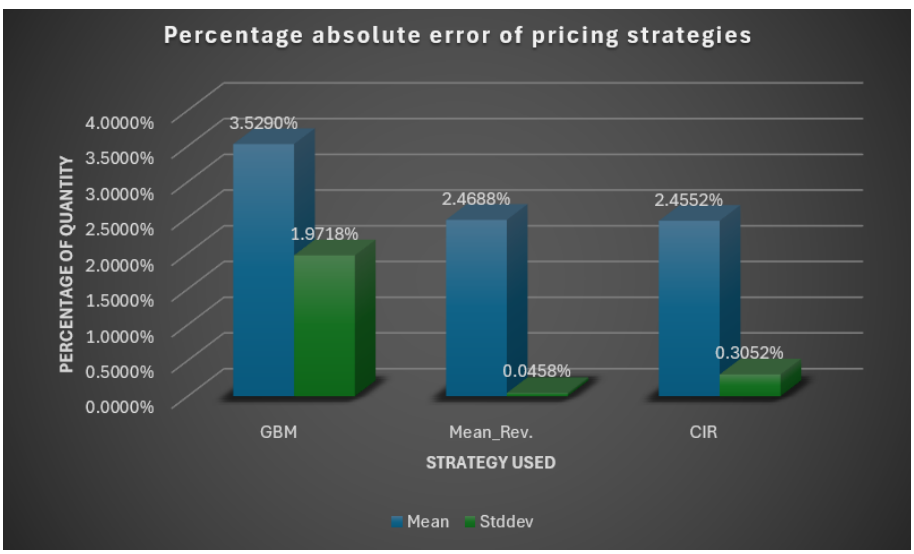

Figure 10: **CTVA** prediction error metrics

6 Discussion

Note: these values might vary slightly from the graph results in the results section. This is attributed to the random input in the shock functions for each model and is normal and expected.

One thing to consider with regards to these results is the fact that historical data can cause overfitting across backtesting or predictive models, which might show up as an unusually low mean error and standard deviation. For the time being, this problem is left for later, as checking for overfitting in Excel is currently out of scope. Instead, the

naïve approach is temporarily adopted: smaller values for mean error and standard deviation always imply a better strategy.

GBM performed the worst consistently across both sectors tested. This is expected given its relatively simplistic assumptions: constant volatility and normally distributed returns. These do not adequately capture the more nuanced dynamics present in real markets. While it remains a useful baseline, it is clearly outperformed by both OUMR and CIR in terms of mean percentage error across the stocks tested.

OUMR and CIR produced approximately the same mean percentage error, making them broadly comparable in accuracy. The key distinguishing factor was standard deviation: CIR exhibited marginally higher standard deviation than OUMR, suggesting its predictions are slightly less consistent from run to run, even when equally accurate on average, which is a trade-off for its ability to manage volatility shock sizes to prevent stock prices from going negative. Although all three models performed horribly for commodities in terms of mean error, OUMR showed a particularly low standard deviation for commodity-based stocks, which is expected given its mean-reverting nature aligns well with the behaviour of commodity prices. CIR performed strongly for blue-chip tech stocks, where its non-negativity constraint and price-proportional volatility better capture the dynamics of high-value equities.

Another thing to note is that only two years of historical data does not provide a holistic insight into the performance of companies. It also does not help to make a versatile model, as that would require more continuous market data over the span of many years to account for minor regime changes, allowing the model to continue performing for longer.

7 Conclusion

This paper compared three stochastic models (GBM, OUMR, and CIR) for short-term stock price prediction across blue-chip tech and commodity-based stocks. The Monte Carlo simulation framework allowed the mean error and standard deviation of each model to be measured consistently across 500 runs per stock, providing a reliable basis for comparison.

GBM performed the worst consistently across both sectors. Its assumption of normally distributed returns and constant volatility limits its accuracy relative to the other two models, and while its computational simplicity makes it a useful baseline, it is clearly outperformed by both OUMR and CIR in terms of mean percentage error.

OUMR and CIR produced approximately the same mean percentage error across the stocks tested, making them broadly comparable in terms of accuracy. The key distinguishing factor between the two was standard deviation: CIR exhibited marginally higher standard deviation than OUMR, suggesting that while CIR is no less accurate on average, its predictions are slightly less consistent from run to run. OUMR showed a particular advantage for commodity-based stocks, which is expected given that its mean-reverting nature aligns well with the behaviour of commodity prices. CIR, meanwhile, performed strongly for blue-chip tech stocks, where its non-negativity

constraint and price-proportional volatility better capture the dynamics of high-value equities.

The key limitation of this investigation is the use of only two years of historical data, which introduces the risk of overfitting to a specific market regime. Future work will extend this analysis to include GARCH(1,1) and its variants, Variance Gamma, and the Heston Stochastic Volatility model, as well as a broader and longer dataset to improve the robustness of the conclusions.

8 Instructions for use

All necessary inputs are yellow cells with red text. For example, A1 is the ticker of the stock that one wants to analyse.

(a) Input the ticker of the stock you wish to analyse in cell A1. Remember that the `stockhistory()` function works for NYSE in default, and can be modified to include other stock exchange, such as the ASX e.g.: XASX.BHP .

(b) The initial values of the graphs should all be 0 when one changes the ticker. This is normal. Head to Formulas → Calculation and click "Calculate Sheet".

(c) You will then have data from those 500 runs that will update on the graphs.

(d) If the standard deviation or mean error for any model is very high, recalculate the sheet. This can be a characteristic of the model or a sign that the ticker entered is not supported by the `stockhistory()` function.

IMPORTANT NOTICE

Keep in mind that the `stockhistory()` function has to be working on your version of Excel, or else the data will not load and the model, graphs and simulations will not work.